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Recovering Distributed LLM Teams After Information Changes

A paired experimental study of conversation retention and recovery

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Recovering Distributed LLM Teams After Information Changes

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1. Abstract

Teams of language-model agents may need to revise their work when information changes. Retaining the previous conversation preserves useful evidence but also potentially obsolete conclusions; clearing it removes both. This study examines that trade-off with two agents holding private lists of integers. After their initial collaboration, the uniquely largest record is replaced with a lower value. Independent continuations then retain or clear the previous visible history, while receiving the same current private data and change notices.

A preliminary experiment exposed weak initial collaboration and frequent communication-rule violations. A subsequent literature review informed a final assisted experiment with clearer instructions, a short coordination strategy, and improved information presentation. Using Qwen3.5-9B, twenty paired main trials produced exact team recovery in 2/20 retain trials (10%) and 3/20 clear trials (15%). The clear-minus-retain difference was +5 percentage points, with a prespecified exploratory 95% bootstrap interval of −15 to +25 points. Neither policy was reliably successful.

Initial collaboration and communication compliance improved descriptively, but different inputs and multiple methodological changes prevent attributing this to assistance. Nevertheless, 32 of 35 failed final recoveries used only legal actions. Recorded interactions show correct reports repeated without final submissions, current values attributed to the wrong source, and correct explanations without committed answers. Improved initial collaboration and more compliant communication did not ensure reliable recovery after information changed.

Keywords: large language models; multi-agent systems; information revision; recovery; conversation history; distributed coordination.

2. Introduction

2.1 Motivation

Collaborative work depends on information that can change. A team may discover that a measurement was wrong, a document was corrected, or an option is no longer available. The problem is then larger than replacing one fact: earlier conclusions may depend on it, while other parts of the work remain useful. Restarting everything wastes valid work. Continuing without reconsideration risks carrying an obsolete conclusion into the final decision.

This problem is especially difficult for teams of large language model (LLM) agents. Their communication is ordinary text. A single message may combine an observation, a calculation, an interpretation, and a proposed action. Another agent can use that message without retaining a precise account of which source supports which conclusion. Updating the original source therefore does not guarantee that the team's later behavior will change appropriately.

Consider two agents searching for the largest value in separate lists. One reports 943 and the other reports 922, so both answer 943. If the record containing 943 is subsequently replaced, the second agent's report may still be useful. The first agent's old maximum is no longer sufficient: it must inspect its current list to find the best remaining value. Keeping the exchange offers a possible shortcut, but also leaves the previous answer visible. Clearing it removes the old answer and the useful report together. This simple example captures the central tension examined here.

The task is adapted from the Global Maximum family in SILO-BENCH, a benchmark of distributed LLM coordination under private information [1]. The present study adds an actual source revision and paired recovery continuations. It uses its own communication rules and exact team endpoint, so the results are not native SILO-BENCH scores.

2.2 Research question and contribution

The research question is:

> *After the previously winning source record changes, how does retaining rather than clearing the agents' previous visible history affect exact team recovery under one fixed, assisted two-agent environment?*

The experiment compares two practical history policies. Retention preserves everything the agents were previously allowed to see, including inaccurate claims. Clearing removes that history. Both policies keep current private sources available and receive the same assistance and change information. The comparison concerns the complete consequences of these policies, including differences in available evidence, wording, and context length. It does not isolate a pure memory mechanism.

The contribution is an exact, traceable account of a small recovery experiment. It combines paired success outcomes with computational effort and concrete interaction examples. In particular, it distinguishes a legal message from a correct claim, a correct claim from a submitted answer, and one correct answer from a successful team. These distinctions matter because a fluent conversation can appear productive while the required joint decision remains wrong or unfinished.

2.3 From the preliminary experiment to the final study

The preliminary experiment tested whether a bounded retain-versus-clear comparison could be carried out with naturally generated collaboration. It used the same nominal model checkpoint, two-agent maximum task, and winning-record withdrawal. Twenty main pairs were completed: each policy succeeded on 2/20 worlds, while only 3/20 original teams were correct. Across pilots and main trials, 60 of 206 team actions violated the communication rules despite having valid JSON syntax. For example, combining a useful message with an incompatible waiting action caused the whole action to be rejected and nothing to be delivered.

The traces also showed numerical errors, confusion between record locations and values, and submissions that contradicted the accompanying explanation. Retain repeated the withdrawn maximum in ten main worlds. An independent audit found no scoring, delivery, or branch-isolation defect explaining these outcomes. The result was usable, but difficult to interpret as recovery from effective collaboration when most original teams had not solved the task.

The subsequent literature reviews connected these problems to interface design, structured actions, memory, and evaluation. They informed clearer action rules, explicit both-agent submission instructions, current/history boundaries, and a short strategy for exchanging and comparing local maxima. The final design also introduced focused diagnostics, four complete pilots, fresh inputs, and a schedule separating device assignment from policy order.

These changes created a new behavioral condition and required a new experiment. Old replies were not repaired or reused under the revised instructions. All ordinary failures remained in the final fixed cohort. Because inputs, presentation, development checks, and scheduling changed together, the preliminary experiment is a developmental comparison, not a controlled test of assistance.

2.4 Scope

The broader project considered more agents, several algorithmic tasks, and more selective recovery policies. This report concerns the executed course study: two agents, one task, one announced source change, and two history policies. The numerical task makes correctness unambiguous and keeps the interactions inspectable. It does not establish performance on realistic research workflows, larger teams, or other models, nor does it introduce a new memory-repair architecture.

3. Related Work

3.1 Revising conclusions while preserving useful work

Doyle's truth maintenance system records reasons for beliefs so that conclusions can be reconsidered when assumptions change [2]. The AGM account formalizes contraction and revision of a theory [3]. Both motivate distinguishing a conclusion from its support, without implying that an LLM's explanation supplies trustworthy dependencies.

DBSP updates computations expressed in a stream algebra [4], while *Build Systems à la Carte* separates computation order from deciding which results need rebuilding [5]. Reuse in these settings depends on defined relationships between inputs and outputs. A free-form peer message does not automatically provide them. This motivates the experiment's separation of current private sources from historical claims.

Chandy and Lamport's snapshot work also explains why communication state belongs in a consistent distributed snapshot [6]. Here, a central scheduler and settled phase boundary permit explicit copies of the original interaction; the study does not introduce a new snapshot algorithm.

3.2 Agent memory and recovery systems

From Faulty Memories to Corrected Actions combines diagnosed faulty memories, provenance, support checks, and selective replay [7]. MEMOREPAIR withdraws affected memory artifacts before publishing validated successors [8]. REVISE uses data and control dependencies to preserve valid workflow progress during revisions [9]. These earlier influences helped delimit the project's claims: selective rollback and source-preserving recovery are established approaches. The present experiment instead leaves agents to interpret uncertified messages, without a host-supplied semantic dependency graph or numerical repair.

StateAuditor distinguishes updating stored information from updating behavior, using previous-to-current transitions to repair responses [10]. Reclaim Evaluation examines memory that preserves a conclusion while losing the source required to recompute it [11]. These distinctions explain why available history need not support successful revision. Unlike Reclaim's setting, our agents retain current private sources and can exchange fresh messages after the edit, so its results do not directly predict our policy contrast.

3.3 Distributed communication and paired evaluation

SILO-BENCH provides the task foundation and distinguishes active communication from effective distributed computation [1]. Our smaller team, source revision, and different waiting, submission, and stopping rules define an additional recovery setting rather than native benchmark scores.

The Causal Intervention Router study reports both rescues and harms when comparing recovery with continued execution from the same state [12]. We likewise report all four paired outcome categories, since an average contrast can conceal opposite changes. Its underlying environment remains unchanged after observation perturbation; ours undergoes an actual source change.

Independent–Communicate–Revise (ICR) fixes initial reasoning and varies communication, measuring both correction and preservation with a no-message control [13]. Its finding that richer communication can increase helpful and harmful influence reinforces the need to examine outcomes rather than message counts alone. ICR agents initially receive complete problems, whereas ours hold private sources.

These papers sharpened the interpretation after the preliminary experiment. They did not originate the project's earlier use of independent continuations.

3.4 Updating information, reasoning, and reliability

Laban and colleagues distinguish strong single-turn aptitude from reliability when information unfolds over several turns [14]. Their findings explain why a successful one-call numerical control is an incomplete readiness test for a team interaction.

LongMemEval separates retrieval, reading, temporal reasoning, and knowledge updates [15]. A visible number can still be read incorrectly, so such an error should not automatically be called forgetting. Belief-R distinguishes revising an answer when evidence requires it from maintaining a still-valid answer [16]. Because our edit always changes the maximum, it cannot test appropriate preservation after harmless changes.

Self-correction and debate results also depend on the comparator. Huang and colleagues find intrinsic reasoning correction unreliable without external feedback [17]. ReConcile provides positive evidence for diverse-model collaboration with explanations, confidence, and aggregation [18]. Neither establishes that extra calls or voting would solve this private-information task with irreversible individual submissions.

3.5 Interfaces, legal actions, and behavioral diagnosis

JSONSchemaBench evaluates structured generation through compliance, constraint coverage, efficiency, and output quality [19]. Tam and colleagues examine the effects of format restrictions on performance [20]. Both support evaluating format separately from task quality. Our runtime validates ordinary generated replies after generation: valid JSON may express an illegal action, and a legal action may contain a false claim.

The refinement drew more directly on interface research. SWE-agent demonstrates that the interface around a fixed model can affect behavior [21]. Workflow assistance in τ²-Bench has different effects across conditions [22], while FlowBench distinguishes choosing an action from supplied history from completing a live interaction [23]. These lessons motivated one transparent assistance package and complete natural pilots, without predicting successful recovery.

CheckList informed focused diagnostics for source reading, instruction following, and commitment [24]. MAST supplies useful descriptions of repetition, termination failures, ignored information, and reasoning/action mismatches [25]. Who&When emphasizes the difficulty of assigning failure to an agent and step [26]. These works guide the case descriptions; they do not make the present AI-assisted review an independent human annotation study or establish causal responsibility.

3.6 How the literature informed the study

The influences occurred in stages: earlier revision and provenance research framed the source/conclusion distinction; preliminary traces exposed practical coordination problems; subsequent reviews informed the assisted interface and live-pilot checks. More selective memory repair remained outside the adopted experiment.

AI Agents That Matter argues for cost-aware evaluation and disclosure of development choices [27]. Accordingly, effort includes failed recoveries and distinguishes physical spending from hypothetical policy cost. τ-bench distinguishes single-run success from consistency across repeated trials [28]. Accordingly, twenty paired worlds under one realized generation schedule provide limited evidence, not repeated-run reliability.

4. System, Methods, and Experiment

4.1 Overall design

Each main trial, called a *world*, contains a fixed pair of private integer lists and one predetermined edit. The agents first collaborate naturally on the original lists. At the end of that phase, the system saves their actual visible histories and submissions. It then creates two independent recovery branches from that same state. One retains history and the other clears it. Both apply the same source change and reopen both agents.

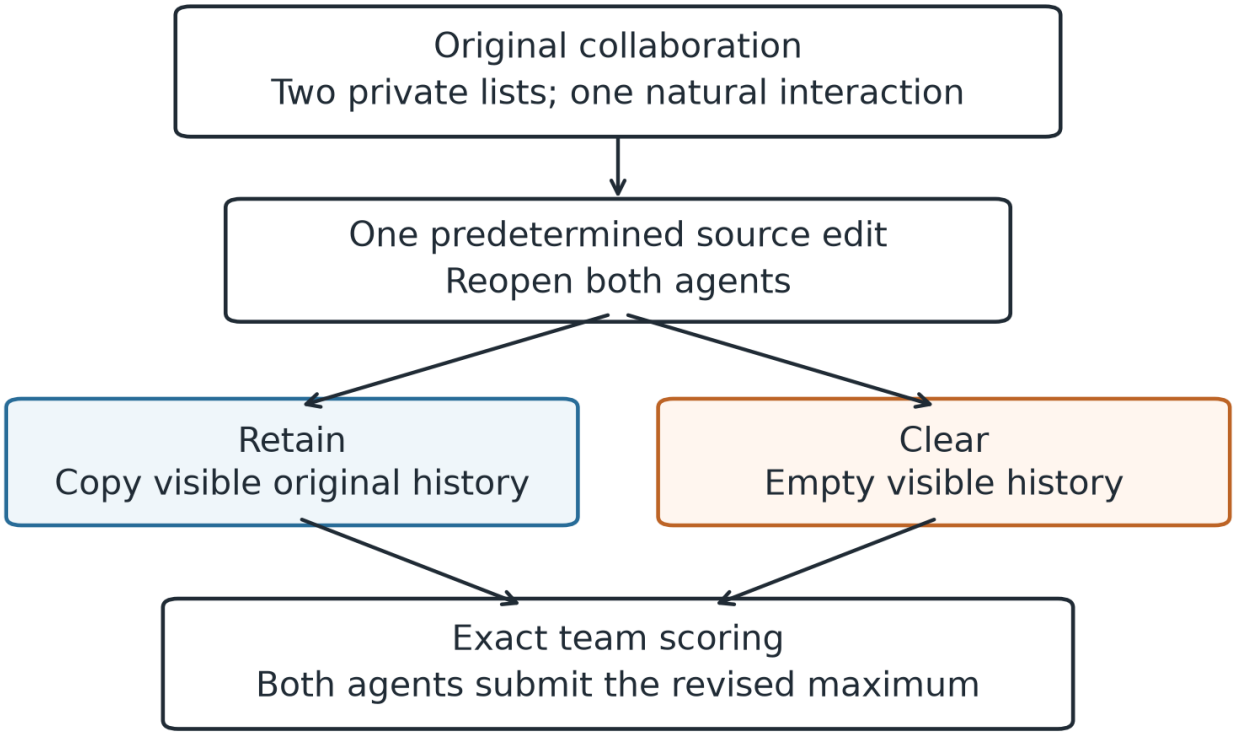


Figure 1. One paired main trial. The original interaction is executed once. Retain and clear are independent continuations after the same edit; the second branch never receives the first branch's messages or answers. A separate full-information control tests maximum finding without distributed interaction.

The pairing controls for the original inputs, edit, and realized initial interaction within a world. It does not make the two recovery requests identical: their histories differ by design. The unit of analysis is the world, rather than an agent, message, or call. The study contains twenty main worlds, plus separate qualification and pilot data.

4.2 Task, input generation, and source change

Each agent receives a private *shard* of twenty integers. Before conditioning, each value is sampled uniformly from −1000 through 1000. Both agents must submit the maximum across all forty values and can communicate only through messages.

An eligible original input has a unique largest record, with value M. Let M_2 be the maximum of the remaining thirty-nine records, with M_2 greater than −1000. The winning record is replaced by

$$x' \sim \text{Uniform}\{-1000, \ldots, M_2 - 1\}, \qquad M' = M_2 < M. \tag{1}$$

The correct answer therefore always changes and decreases. Repeating the old answer cannot solve recovery, while the changed agent's old local maximum need not reveal its best remaining value.

Inputs are conditioned on the scheduled owner of the original winner, with rejection occurring before model outcomes exist. The study contains four qualification, four pilot, and twenty main worlds. All generated original/revised complete inputs are distinct and were checked against locally available historical inputs. This exact-input check does not rule out pretraining contamination.

The main cohort has five worlds in each changed-owner/first-speaker combination. First speaker is fixed across the original and recovery phases within a world. Inputs, edits, assignments, and seeds were saved before inference; no world was selected, replaced, or excluded on the basis of model performance.

4.3 What the agents see

Each call contains common instructions, agent identity and call count, whether the partner has finished, visible history, any change notice, and the current private shard. History precedes current values. The source records are neither sorted nor marked with a maximum.

The fixed assistance explains the global objective, legal actions, source ownership, private submission, waiting, and finality. It suggests establishing both local maxima, comparing them, and communicating useful information before finishing. This supplied strategy means the study does not test autonomous discovery of the maximum algorithm. It is guidance: the host neither forces a fresh exchange nor calculates, certifies, or repairs numerical claims.

After revision, both agents learn the changed owner's identity and the record index. Only the owner sees the replacement value in its notice and current shard. The old value is not supplied again as current evidence. The winning-record eligibility rule, evaluator answers, the other private shard, future edits, and the policy label remain hidden. The common instructions state that only the named record changes, so an accurate old report from the unchanged peer can remain useful under retain.

4.4 Actions, communication, and stopping

An agent returns one JSON object with three fields: `message`, `submit`, and `wait`. A message is text or null, a submission is an integer or null, and waiting is a boolean. The action rules give these fields operational meaning.

Choice	Effect
Send a message	Deliver its text to the partner
Submit an integer	Record a private final answer and finish the sender
Send and submit together	Deliver the text, then finish the sender
Wait alone	Block the sender until a message arrives
Invalid reply or illegal combination	Consume the call, deliver nothing, and record rule feedback

Table 1. Action choices and their effects. Numerical correctness is assessed separately from legality.

Waiting must be the only action. Combining it with a message or submission is illegal, as is doing nothing while not waiting. A legal message can contain an incorrect maximum because feedback checks public action rules, not truth. Rejected text is preserved for audit but is not delivered to the partner.

Each agent has at most three calls in each phase, with no transfer of unused opportunities. Eligible agents act in alternating order. If one has finished or is blocked, the other can act if eligible. Legal messages arrive before the next eligible call. This is a sequential interaction protocol.

Submission is irreversible within the phase. The partner-finished flag reveals that a submission occurred, but not its value. An agent can disclose its answer explicitly in a message, including when sending and submitting together. A phase ends when both agents have submitted, or when waiting or exhausted call

opportunities prevent further progress. Correctness never determines stopping. Thus an incomplete terminal team is a known failure, not an unobserved outcome, and one correct submission does not count as team success.

4.5 History policies and independent recovery

At revision, both agents are reopened with fresh call counters, waiting flags, and submissions. Their current sources and permitted change notices are identical across the two policies.

Retain copies each agent's actual visible original history, including its legal own actions, explicit old submissions, received messages, and public rejection feedback. Incorrect claims remain unchanged. Earlier-phase and current-phase entries have explicit boundaries. Raw rejected replies remain audit records rather than becoming delivered history.

Clear starts with empty visible histories. It keeps the same common instructions, current private source, source-change notice, and fresh phase state as retain. It is still a recovery condition: the agent sees that information changed and that the phase was reopened.

Only the original interaction is shared physically. Each recovery branch has its own messages, submissions, and effects. Per-agent call seeds are matched across policies by call index, but the requests differ because the histories differ. Equal seeds do not guarantee equal outputs, and continuation replies are never reused across policies. Hidden model state is recreated for each call; retention refers to visible history, not a persistent private chain of thought or model cache.

The intervention changes the whole history package. Useful evidence, obsolete conclusions, length, and phrasing vary together. It cannot identify the effect of one old sentence or distinguish a specific internal bias from other historical influences.

4.6 Preparation, pilots, and main execution

Configuration, instructions, inputs, scoring, and admission rules were fixed before the first call. Preparation included state-isolation and hardware checks, eight qualification controls, and four single-call behavioral diagnostics. A *full-information control* receives all forty values and returns a maximum without communication. It is a separate stochastic condition, not an oracle constraining the team.

Diagnostics probed instructions, current-source reading, and supplied-history use. They required settled evidence but had no aggregate accuracy threshold for main admission. Four complete natural pilots then ran the original interaction, both recovery policies, and original/revised controls.

Main admission required complete pilot evidence, healthy execution, sufficient resources, at least 3/4 correct original teams, and at least 7/8 correct controls. Successful recovery or a favorable policy difference was not required. Once admitted, every main world received the original phase, both recoveries, and a revised control. Incorrect original teams remained included.

Recovery order and device assignment were balanced ten-to-ten and varied independently, with five worlds in each joint device/order combination. Both recoveries of a world ran on the same device. This separated the preliminary schedule's perfect device/order coupling without changing its historical records.

4.7 Model, implementation, and resources

The experiment used the pinned Qwen3.5-9B checkpoint [29] on two NVIDIA Tesla T4 devices, with four-bit NF4 double quantization and FP16 computation. Direct generation used temperature 0.7, top-p 0.8, top-k 20, repetition penalty 1, presence penalty 0, and one beam. Input/output limits were 8,192/512 tokens. There was no fine-tuning, thinking-mode fallback, or constrained decoding.

A Python controller recorded each request, raw reply, and resulting state in SQLite. One resident model worker served each device, including both logical agents of its assigned world. Outcomes were saved

before their effects, allowing completed calls to resume without resampling. Each model call used fresh hidden state; only the permitted visible history crossed calls.

All generations, including diagnostics and failures, counted toward a 500-attempt ceiling. Wrong or illegal replies were never retried, and no infrastructure retry occurred. The recorded minimum free-memory headroom was approximately 6.78 GiB, above the 1.5 GiB gate.

Independent offline reconstruction checked inputs, permitted requests, delivered histories, actions, terminal outcomes, and effort against the original records. It establishes consistency of the saved evidence and scoring, not reproduction of GPU arithmetic or independent human annotation.

4.8 Outcomes, effort, and uncertainty

For world w, let R_w and C_w equal one when both agents' final revised submissions equal the exact revised maximum under retain and clear, respectively, and zero for a known terminal failure. An unresolved or unexecuted episode would remain unknown. The primary estimate is

$$\hat{\delta} = \frac{1}{20} \sum_{w=1}^{20} (C_w - R_w). \tag{2}$$

A positive value favors clear on exact team success. Both policy rates and all four paired outcome categories are reported, so opposite changes across worlds remain visible.

The prespecified exploratory uncertainty display uses 10,000 paired bootstrap draws, stratified by changed owner and first speaker. Each draw samples five complete world pairs with replacement within each of the four strata, then averages the twenty paired differences. Exact seeds, ordering, and percentile selection are supplied with the analysis code. The resulting nominal 95% interval is descriptive; its small-sample coverage has not been established. It is neither an equivalence test nor a measure of reliability across repeated generation schedules.

The principal secondary outcome is the mean paired difference in post-revision ready-model inference time, including failed episodes. This is time spent generating with an already loaded model. It excludes loading and is distinct from elapsed notebook time or provider billing. Calls, input/output tokens, delivered messages, completion, wrong agreement, and absent submissions provide additional descriptive outcomes. Each hypothetical end-to-end policy includes the original interaction logically, although that interaction ran only once physically.

5. Results and Discussion

5.1 Coverage and the feasibility findings

All twenty intended main worlds were observed, including each original interaction, both recoveries, and the revised control. There are no unknown main outcomes or effort values. The campaign used 356 physical first attempts, with no retries or contingency charges.

Qualification controls were correct in 8/8 cases. All four behavioral diagnostics produced legal actions, but only one met its exact target, warning of instruction-following, source-reading, and commitment weaknesses. The natural pilots produced 4/4 correct original teams and 8/8 correct controls, yet all eight pilot recoveries failed.

The pilots nevertheless met the fixed admission rule: it tested initial collaboration, numerical capability, complete evidence, and feasibility rather than reliable adaptation. Main execution was not selected for favorable recovery performance, and the candidate was not tuned after the pilot failures.

5.2 Exact main recovery

Retain succeeded on **2/20 worlds (10%)** and clear on **3/20 (15%)**. There were two retain-only successes, three clear-only successes, no shared success, and fifteen worlds where neither policy succeeded.

The three clear-only and two retain-only successes give a clear-minus-retain difference of $(3 - 2)/20 = $ **+5 percentage points**. Its adopted exploratory 95% bootstrap interval is **−15 to +25 percentage points**. Only five worlds distinguish the policies on success; one changed world outcome moves a policy's rate by five points. The interval allows effects favoring either policy, and the data establish neither superiority nor equivalence.

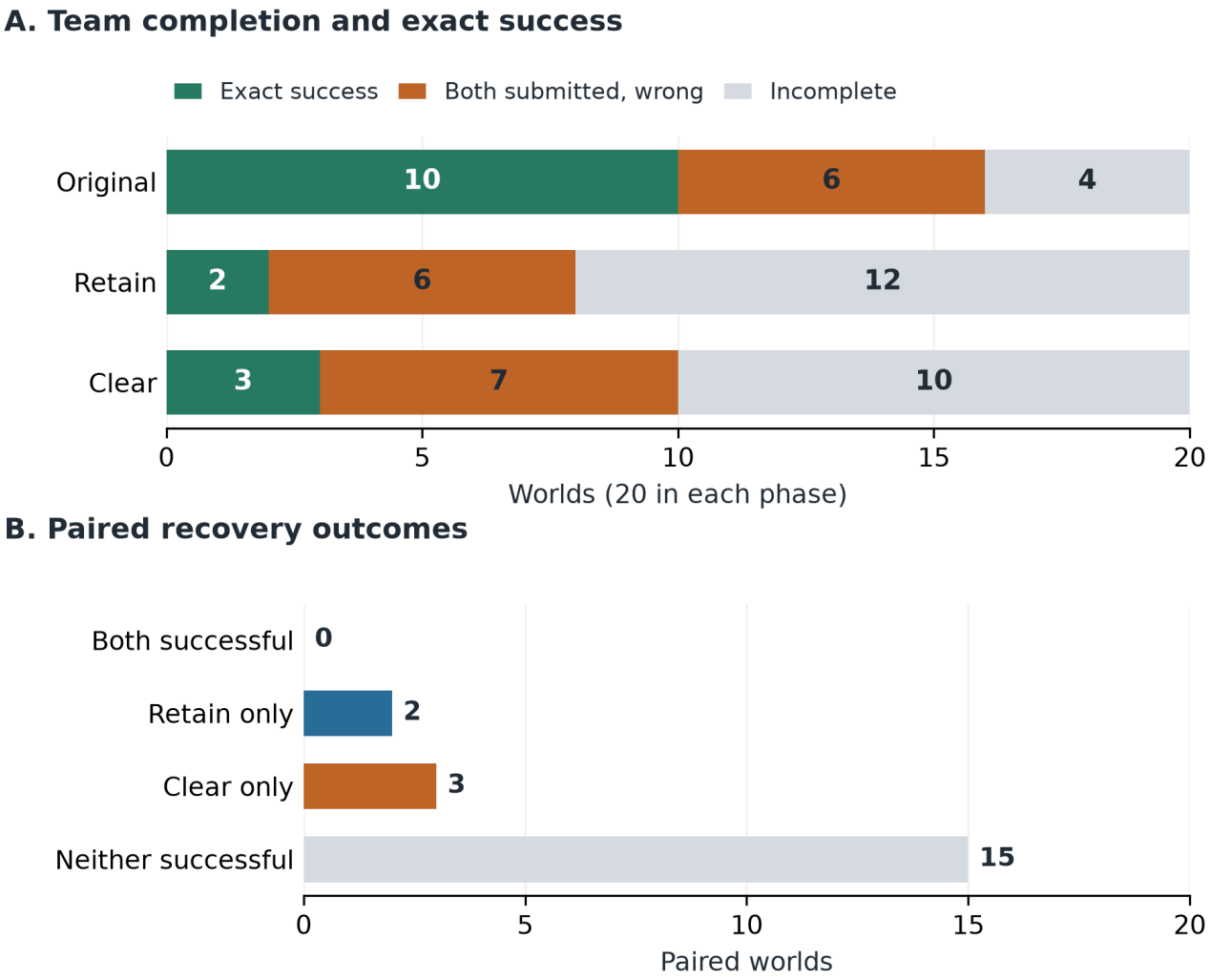


Figure 2. Exact main outcomes. Panel A separates successful, completed but wrong, and incomplete teams; each bar contains twenty worlds. Panel B preserves the pairing of retain and clear. Incomplete terminal episodes are known failures, not missing observations.

The original collaboration was correct on 10/20 worlds. Among those ten, retain recovered on two and clear on one. Clear's other two successes followed incorrect original teams. These are descriptive subset counts, without a separate interval or conditional-effect claim. The primary estimate includes all twenty worlds because selecting only successful original teams would answer a different question.

The revised full-information controls were correct on 18/20 worlds. A control is a separate stochastic call with different instructions, information access, and seed, so it is not a deterministic ceiling on team performance. Conversely, the many correct controls show that the recovery result cannot simply be described as universal inability to find a maximum. Single-call numerical ability and successful distributed recovery are different requirements.

5.3 Compliance, completion, and correctness

All 348 scientific responses had valid JSON syntax and the required field types. Six combined a message with an incompatible waiting action and were rejected. All six occurred in main natural episodes; every pilot team action was legal. Across pilots and main collaboration, 302 of 308 team actions were legal.

Main-phase measure	Original	Retain	Clear
Calls	82	91	85
Illegal actions	1	1	4
Delivered messages	70	85	77
Both agents submitted, out of 20 teams	16	8	10
Exact success, out of 20 teams	10	2	3
Correct final answers, out of 40 agent slots	30	14	18
Wrong final answers, out of 40 agent slots	6	8	11
Known absent answers, out of 40 agent slots	4	18	11

Table 2. Main protocol and submission coverage. Calls depend on behavior; team and final-answer denominators remain fixed. There are no unknown slots.

Retain's eighteen failed recoveries consist of twelve incomplete teams and six completed but wrong teams. Clear's seventeen failures consist of ten incomplete teams and seven completed but wrong teams. Crucially, **17/18 retain failures and 15/17 clear failures contain no illegal action**. Thus 32 of 35 failed recoveries obey the action rules throughout. This does not identify a causal fraction of failures, but rejected actions do not describe the dominant observed pattern.

Retain delivered more messages yet completed fewer teams. Counting exact repeated strings beyond each sender's distinct messages within an episode gives thirty-one duplicates under retain and eighteen under clear. Thirteen retained agent trajectories repeated the same message on all three calls, compared with four clear trajectories. These are literal string counts, not a complete detector of conversational loops. Repetition can sometimes be useful, while changed wording can conceal the same stalled decision.

No final recovery submission in either policy repeated the withdrawn global maximum. This differs from the preliminary experiment, but does not show that history had no influence. History can affect behavior without reproducing the old number. Moreover, clearing removes that number from both the current data and previous visible conversation, making old-value availability unequal by construction.

5.4 What happened in the interactions?

World numbers refer to Appendix A. Worlds 00 and 01 are the prespecified first clear-only and retain-only successes; other examples were selected retrospectively to illustrate distinct observed failures. These cases are illustrations, not an independently coded causal taxonomy.

Correct information without a final decision

World 08 had a correct original answer of 874. After the winning record changed, the actual local maxima were 758 and 857, making 857 the revised global answer. Under retain, both agents correctly reported their current maxima, then repeated the same messages on their remaining calls.

Agent	Delivered message, repeated on all three calls	Final answer
0	"My local maximum is 758 at record 19. I will wait for your maximum before submitting."	Absent
1	"My local maximum is 857 at record 12. I will wait for your maximum before submitting."	Absent

Table 3. A legal report loop in world 08. Each message arrived before the partner's next eligible call.

Both values needed for the comparison were therefore present in the conversation, but neither agent submitted. The phrases about waiting were only message text: the actual waiting fields were false. The team exhausted six calls while continuing to communicate. This was neither lost delivery nor a protocol deadlock.

Clear failed differently on the same world. Agent 1 reported 786 despite the visible 857 in its private list, and both agents finally submitted 786. That branch completed and agreed, but was wrong. Together, the two branches distinguish accurate communication without completion from completed agreement based on inaccurate information.

A correct explanation with no committed answer

In world 04 under clear, agent 0 sent its correct maximum, 954, and later submitted it. Agent 1's last message correctly compared its own 839 with the partner's 954 and stated: "954 is the global max. I will submit." Its actual submission field remained null. The partner had finished, and no further opportunity remained for agent 1, so the team failed.

The observable problem is precise: the right value appeared in a delivered report and in the agent's own explanation, but not in its final action. Scoring the prose as an answer would repair the experiment retrospectively. Explanations and committed decisions must therefore be evaluated separately, as the failure-analysis literature recommends [25].

Confusing a record's owner with its location

In world 09 under clear, agent 1 received a current message saying that agent 0's maximum was 949 at record 16. The source notice identified agent 0's changed record as 13. Agent 1 then claimed that the partner's maximum was 695 at record 13 and submitted its own 836. The number 695 was present at index 13 of **agent 1's own list**. The correct 949 report was visible in that same request.

The visible correspondence supports a description of source-ownership confusion: a remote record location was associated with a locally visible value. It does not reveal the internal cause. The example occurs with cleared history, so obsolete retained text is not necessary for this behavior. LongMemEval's distinction between available information and successful reading is relevant here [15].

Success under either policy, with limits on the explanation

World 00 was clear-only successful. The original answer was 943 and the revised answer 922. Clear exchanged the correct current maxima, 900 and 922, and both agents submitted 922. Retain instead reported 837 for the changed shard despite its visible 900, repeated messages, and ended with no submissions. The difference combines source-reading and completion behavior; it does not isolate a harmful historical sentence.

World 01 was retain-only successful. An unchanged-peer maximum of 926 was present in the old history and was also reported afresh. The changed owner established its current maximum of 931, compared the two, and submitted 931; the partner followed. Clear exchanged the same correct maxima, but only one agent submitted, while the other continued requesting information and finally waited. Retain's success is consistent with useful reuse of unchanged evidence, but the fresh report prevents attributing the outcome uniquely to historical memory.

Correct outcomes can also contain incorrect premises. Clear solved world 19 in two calls with both submissions equal to 896. Its first speaker nevertheless invented a remote maximum of 639, a value in its own shard; the true remote maximum was 860. Since 896 exceeded both, the false premise did not change the answer. Exact scoring properly credits these outcomes, while leaving faithful source tracking and robustness unproven.

Finishing too early and failing to finish

In world 17 under retain, an agent irreversibly submitted 663 despite having 819 in its current private list. Later information could not reopen that commitment. In world 11 under retain, one agent correctly submitted 950 and the other subsequently chose a legal wait. The finished partner could no longer send a message to wake it, so the phase deadlocked even though only one agent was waiting.

The opposing demands are to obtain enough evidence before committing and to finish once it is available. The traces do not establish whether more turns or reversible submissions would rescue either case.

5.5 Why improved collaboration did not produce reliable recovery

Recovery required current-value reading, source ownership, interpretation of changed information, uptake of peer reports, and final commitment to succeed together. The cases show correct information without decisions, delivered evidence replaced by an unrelated local value, and a correct explanation without a submission. These are more precise observations than describing every failure as repetition of an obsolete answer.

The source/conclusion distinction in Reclaim remains useful [11], but preserving or reacquiring the right content was not always sufficient to produce action. Retained text might encourage repetition of an ineffective pattern; this remains a hypothesis, not an identified internal cause.

Recovery also introduces reopening and a notice, even when history is cleared. Without a matched no-change reopening, the decline from 50% original success to 10–15% recovery cannot be assigned solely to stale history or to factual revision itself. Likewise, the full-information controls change communication, information access, and commitment requirements together; their higher accuracy does not identify a causal communication-failure rate.

5.6 Recovery effort

Measure, mean per main recovery	Retain	Clear
Model calls	4.55	4.25
Ready-model inference time	29.439 s	25.656 s
Input tokens	5,146.10	4,017.60
Generated tokens	170.55	159.70
Delivered messages	4.25	3.85

Table 4. Post-revision effort across twenty worlds, including failures. Time contrasts use unrounded records.

Clear used an average of **3.783 fewer ready-model seconds per world**. The paired clear-minus-retain effort interval was **−8.241 to +0.589 seconds**, including zero. Mean paired differences were −0.30 calls, −1,128.50 input tokens, −10.85 generated tokens, and −0.40 delivered messages per world. No general efficiency advantage is established.

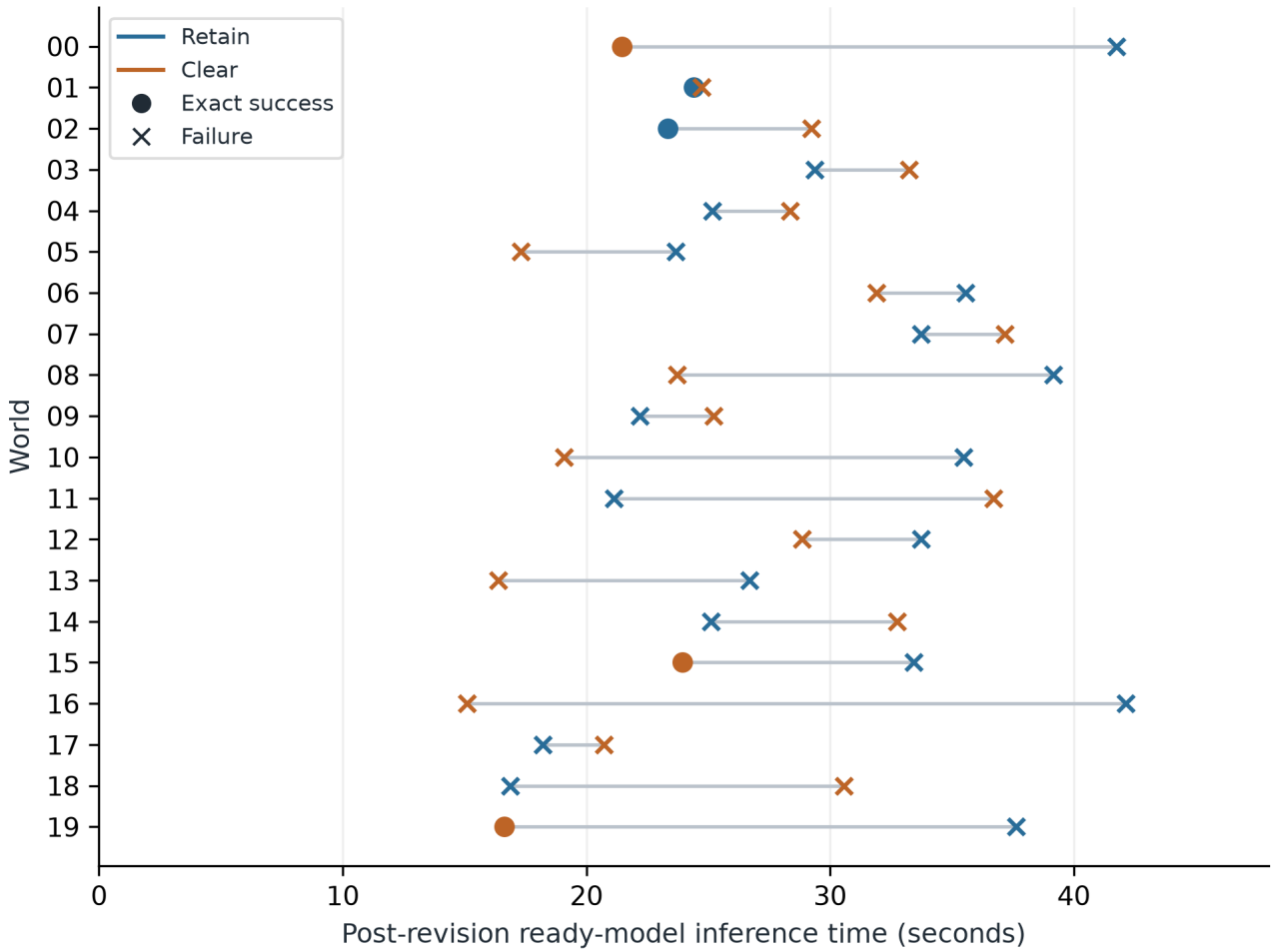


Figure 3. Ready-model inference time for each main world's paired recoveries. Lines connect the same world; circles mark exact success and crosses mark known failure. Both agents' calls are included. A shorter recovery may reflect an early wrong commitment or deadlock.

Fast failure is an important qualification. In world 16, both branches ended with the same wrong disagreement, [601,500]. Clear took about 15.11 seconds and retain 42.14 seconds. In world 11, retain was cheaper because it deadlocked early, while clear continued. Time savings should therefore be read alongside success and termination.

The shared original interactions cost eighty-two physical calls and 493.775 ready-model seconds. Charging them to each hypothetical complete policy gives 173 logical calls and 1,082.553 seconds for retain, versus 167 calls and 1,006.890 seconds for clear. Adding those two totals would double-count the shared original interaction. Controls and hardware checks are research overhead, rather than spending attributable to either distributed policy.

The observed campaign lasted approximately 38.26 minutes. Summed generation time differs from elapsed time because loading and orchestration add overhead while work can overlap. Neither is a measure of provider billing or energy use. All scientific replies ended normally within the token limits; the longest used 124 of the 512 permitted output tokens. Truncation does not explain the observed failures.

5.7 What changed relative to the preliminary experiment?

Observed measure	Preliminary experiment	Final assisted experiment
Original team success	3/20	10/20
Retain recovery success	2/20	2/20
Clear recovery success	2/20	3/20
Clear-minus-retain difference	0 points	+5 points
Exploratory success interval	[−10,+10] points	[−15,+25] points
Revised full-information controls correct	15/20	18/20
Illegal actions, all pilot/main team calls	60/206	6/308
Retain teams with both submissions	20/20	8/20
Clear teams with both submissions	19/20	10/20
Retain worlds submitting the obsolete answer	10/20	0/20

Table 5. Descriptive comparison of separate experiments. Inputs are different, and identically numbered worlds are not matched units. The preliminary interval retains its original analysis convention.

The final experiment exhibits a thirty-five-point higher initial success rate, many fewer illegal actions, and no obsolete recovery submissions. Within the main original interactions, illegal actions numbered 42/82 in the preliminary experiment and 1/82 in the final experiment. These differences are meaningful observations. They do not establish that assistance caused the improvements. Common instructions, action examples, source placement, history boundaries, feedback, input cohort, seeds, qualification/pilot stages, and execution assignments changed together. There is no same-world comparison with assistance turned off.

The preliminary experiment also coupled device and policy order perfectly. The final schedule separated them and kept paired recoveries on the same device. That improves the balance of the final study; it does not retrospectively make the preliminary cohort a randomized control.

The behavioral change is still informative: preliminary retain often finished with an obsolete answer, while final retain more often communicated without finishing. Retain's recovery calls increased from forty-four to ninety-one, but successful teams remained two. Better collaboration and compliance therefore coexisted with poor recovery; the executions should not be pooled or interpreted as an isolated assistance effect.

5.8 Limitations

Sample and replication. Twenty worlds and one realized generation schedule provide low precision, particularly with only five discordant success pairs. The bootstrap describes empirical world variation; its small-sample coverage is unestablished and it does not measure stability over repeated executions.

Intervention and controls. Retain and clear differ in useful evidence, obsolete content, context length, and wording. Half the original teams were incorrect, so the primary result concerns natural collaboration rather than recovery only from correct prior work. There is no matched no-change reopening or answer-preserving edit. The study cannot isolate revision-specific disruption or establish appropriate preservation of still-correct answers [16].

Generalization. Results concern one quantized model, two agents, a short call budget, and an edit that always lowers a unique maximum. The task contains little expensive computation to preserve. It informs behavior after changed information more directly than savings from computation reuse, and does not establish performance on larger teams or professional workflows.

Behavioral interpretation. Qualitative review was AI-assisted and unblinded, without independent human coding, inter-rater agreement, or counterfactual repair tests. Exact traces establish what was visible and what happened, not private cognition or which single correction would rescue a failure [26].

Runtime. Offline reconstruction checks the saved experiment. No crash, retry, or cross-session restoration was observed, and GPU numerical equivalence was not independently reproduced. The findings do not certify every hidden backend state.

5.9 Implications and future work

The immediate priority is to test the complete decision process: current-value extraction, source ownership, uptake of a delivered report, and final commitment after partner completion. Focused tests, as suggested by CheckList [24], should be followed by live interaction, since using a supplied correct history differs from producing it.

A matched no-change reopening would help separate factual revision from generic reopening and interface effects. An answer-preserving edit would test whether valid conclusions survive harmless changes. To study historical influence more narrowly, a further condition could hold useful evidence fixed while varying exposure to the old conclusion [11], [13]. Each addresses a different question from the present natural-history comparison.

Fresh worlds and repeated continuations are needed to assess stability. Proposed changes to models, call limits, communication, or feedback should be evaluated prospectively with their costs, rather than used to repair the recorded outcomes.

6. Conclusion

This study compared retaining and clearing a two-agent LLM team's visible history after an actual source change. In the final assisted experiment, all twenty intended main world pairs were observed. Retain achieved exact team recovery in 10% of worlds and clear in 15%. The paired difference of +5 percentage points had an exploratory interval from −15 to +25 points, providing no basis for a general preference between the policies.

The stronger finding concerns the difference between communication and completed action. Initial collaboration was more successful and illegal actions were much less frequent than in the preliminary experiment, although the comparison cannot identify assistance as the cause. Recovery remained poor under both policies. Correct reports sometimes circulated without submissions, current values were misread or attributed to the wrong source, and stated conclusions did not always match final actions. Even some successful decisions contained false intermediate claims.

The contribution is a bounded negative result supported by complete paired outcomes, exact scoring, computational effort, and inspectable interactions. Preserving evidence differs from preserving conclusions, and making information available differs from using it successfully. In this setting, improved initial collaboration and compliant communication did not ensure that both agents reconsidered current evidence and finished correctly after information changed.

Code and data availability. The private companion repository, *llm-team-recovery-study*, supplies the executed notebooks, unchanged Kaggle downloads, experiment source, configuration, and offline analysis for both experiments. Readers require repository access. Exact inputs, prompts, seeds, runtime versions, and trace locators are retained there rather than reproduced in this report. The historical prose ledger misprints 354 of 356 per-call seed labels; the saved requests contain the exact integers used in reconstruction, and the outcomes and statistics are unaffected.

Assistance disclosure. Drafting, editorial revision, and qualitative trace review used AI assistance. Numerical checks do not replace independent human behavioral annotation.

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Appendix A. Complete main outcomes

Submissions are ordered [agent 0, agent 1]. A dash denotes a known absent answer in a terminal episode. **B** means both agents submitted; **H** means the call horizon was exhausted; **W** means waiting deadlock. These end reasons do not imply correctness. The world numbers refer only to the final assisted experiment.

World	Old → revised maximum	Original answers	Retain answers / end	Clear answers / end
00	943 → 922	[943,943]	[—,—] H	[922,922] B
01	956 → 931	[956,956]	[931,931] B	[—,931] H
02	976 → 964	[976,976]	[964,964] B	[964,—] H
03	946 → 893	[—,946]	[810,893] B	[876,—] H
04	972 → 954	[972,972]	[—,839] H	[954,—] H
05	952 → 884	[952,868]	[884,—] H	[884,868] B
06	941 → 917	[941,941]	[917,—] H	[766,—] H
07	991 → 980	[991,991]	[—,—] H	[—,980] H
08	874 → 857	[874,874]	[—,—] H	[786,786] B
09	979 → 949	[979,836]	[949,836] B	[949,836] B
10	991 → 987	[878,991]	[—,987] H	[878,987] B
11	984 → 950	[—,984]	[—,950] W	[950,—] H
12	963 → 905	[963,—]	[—,—] H	[—,—] H
13	992 → 873	[992,992]	[873,864] B	[873,864] B
14	980 → 975	[980,980]	[975,—] H	[975,—] H
15	917 → 815	[815,917]	[—,—] H	[815,815] B
16	702 → 601	[702,—]	[601,500] B	[601,500] B
17	974 → 955	[974,955]	[663,955] B	[—,955] H
18	961 → 912	[961,961]	[805,805] B	[764,805] B
19	931 → 896	[860,931]	[—,—] H	[896,896] B

Table A1. Every intended main world. No absent answer or unfavorable outcome is excluded. Revised full-information controls matched the revised maximum in every row except world 15 (881 instead of 815) and world 16 (585 instead of 601).